A Note on Reciprocal-Chart Folding over Quartic Extensions

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Abstract

We study reciprocal-chart folding over a quartic extension E/F. Fix a seed $\tau \in E$ with $F(\tau) = E$, and let the public orbit be defined by

$$r_1 = \tau, \qquad r_{i+1} = \frac{1}{r_i - c_i}, \qquad c_i = \begin{cases} 0, & i \text{ odd}, \\ 1, & i \text{ even}. \end{cases}$$

For F-valued leaves, we show that every round admits an exact reciprocal transport in the moving basis $\{1, r_i, r_i^2, r_i^3\}$. This yields, for every depth $n \geq 1$, a seed-specialized public F-linear evaluator

$$T_{\tau,n} : F^{2^n} \to F^4$$

whose live state always has width four over F.

Our main theorem gives an exact descriptor for this evaluator family. From the seed descriptor (n, τ), one can compute in $O(n)$ base-field operations a reduced fold descriptor of size exactly $4(n-1)$ field words together with 8 field words of reconstruction data. The reduced descriptor supports $O(n)$-time coefficient queries for entries of the matrix of $T_{\tau,n}$, and the physical fold engine evaluates $T_{\tau,n}(x)$ in $O(N)$ time for $N = 2^n$. Under the canonical straight-line implementation analyzed here, the fold engine performs exactly $2N - 4$ base-field multiplications on F-valued leaves.

The note isolates a cryptographically relevant algebraic backend for downstream commitment/opening protocols and non-interactive proof compilers: a constant-width seed-specialized evaluator, its exact descriptor size, and the interface consequence that a downstream layer restricted to explicit row vectors cannot preserve the $O(n)$-word specialization. A minimal information-theoretic oracle-model one-shot soundness theorem for the evaluator relation under pre-committed outputs and fresh verifier randomness is also established.

1 Introduction

Reciprocal-chart folding appears naturally in sumcheck-style folding questions [12, 4]. Once a post-commit seed τ is fixed, however, the cleanest object is not an interactive transcript but the induced linear evaluator

$$T_{\tau,n} : F^{2^n} \to F^4,$$

obtained by iterating a public Möbius orbit and an exact round-by-round reciprocal transport. The purpose of this note is to isolate that evaluator family in the first nontrivial closed-form case: a quartic extension E/F with the alternating 0/1 orbit.

Existing presentations of sumcheck-style folding [12, 4] and related algebraic backends typically describe the round-by-round update at a schematic level, leaving the concrete transport identities, descriptor sizes, and arithmetic costs implicit. In particular, for extensions of degree greater than two, no prior work isolates an exact closed-form local fold identity in a moving power basis, a

fully explicit reduced descriptor with a provable size bound, or the precise multiplication count of the resulting engine. This gap matters for downstream layers: without an exact algebraic specification, commitment/opening protocols and non-interactive compilers must either re-derive the round structure internally or fall back to generic interfaces that forfeit the logarithmic-size seed specialization.

This backend viewpoint is cryptographically useful for two complementary reasons. First, commitment and opening layers for structured linear families must decide whether they can consume a seed-dependent structured descriptor or whether they must materialize explicit rows [5, 8, 1, 9]. Second, later non-interactive or correlated-challenge analyses are cleaner once the algebraic backend has already been stated as a precise linear object [7, 3]. In the quartic reciprocal chart studied here, that backend is especially rigid: the live state stays width four over F, the seed specialization is describable in $O(n)$ words, and explicit-row interfaces lose that succinctness.

Contributions. The note makes the following concrete claims.

1. We prove an *exact local reciprocal fold identity* in a moving quartic basis.
2. We prove a *structured evaluator theorem*: for F-valued leaves, the seed-specialized map $T_{\tau,n}$ admits a reduced public descriptor of size exactly $4(n-1)$, an $O(n)$-time coefficient oracle, and an $O(N)$-time evaluation procedure.
3. We prove an *exact canonical multiplication count* of $2N-4$ for the fold engine on F-valued leaves.
4. We record an *interface consequence*: any downstream layer that insists on explicit row vectors must materialize the four rows of $T_{\tau,n}$, and therefore cannot preserve the $O(n)$-word seed-dependent specialization.

Related work and positioning

Classical sumcheck originates with Lund, Fortnow, Karloff, and Nisan [12]; modern cryptographic presentations include [4], and recent implementation-oriented analyses include [2, 13]. On the commitment side, succinct openings for structured linear families are studied by, among others, [5, 8, 1, 9]. On the folding side, recent protocol-level systems such as Nova, ProtoStar, and HyperNova develop recursive or accumulation-based proof layers above algebraic backends [11, 6, 10]. The present paper isolates a seed-specialized linear evaluator and records a minimal oracle-model one-shot soundness statement that later cryptographic layers may refine.

Scope and organization

We focus on the quartic case $[E:F]=4$ and on the alternating 0/1 schedule because this is the first degree in which the moving basis, reciprocal transport, coefficient update, and exact arithmetic constants all fit in a short self-contained note. The aim is precision rather than maximal generality. Sections 2–4 develop the algebraic backend, Section 5 gives the worked $N=4$ example, Section 6 fixes the arithmetic cost model, Section 7 establishes the explicit-row interface consequence, and Section 8 offers concluding remarks. The appendix establishes a minimal information-theoretic soundness statement in an idealized oracle model with pre-committed outputs; concrete commitment/opening and Fiat–Shamir layers build on this foundation as subsequent work.

2 Algebraic Setup and Reciprocal Orbit

Sections 2 through 5 develop the algebraic backend, with Section 5 giving the first concrete worked example. Section 6 then fixes an arithmetic cost model in order to state exact multiplication counts. Let F be a field of characteristic different from 2 and 3, let E/F be a quartic extension, and let

$$N = 2^n, \qquad n \geq 1.$$

We index leaves by

$$j \in \{0, \ldots, N-1\}, \qquad j = \sum_{i=1}^{n} b_i 2^{i-1}, \qquad b_i \in \{0, 1\},$$

using the least-significant-first path convention.

Whenever an integer k appears in a field expression, it denotes its image in F.

Define the algebraic seed domain

$$\Omega^{\mathrm{alg}} := \{\tau \in E : F(\tau) = E\}.$$

For $\tau \in \Omega^{\mathrm{alg}}$, define the public challenge orbit by

$$r_1 = \tau, \qquad r_{i+1} = \frac{1}{r_i - c_i}, \qquad c_i = \begin{cases} 0, & i \text{ odd}, \\ 1, & i \text{ even}. \end{cases}$$

Lemma 2.1 (Orbit formulas and moving charts)**.** *Let $\tau \in \Omega^{\mathrm{alg}}$. Then every reciprocal update is well defined. Moreover, for every admissible k one has*

$$r_{2k+1} = \frac{\tau}{1 - k\tau}, \qquad r_{2k+2} = \frac{1 - k\tau}{\tau},$$

and for every admissible round i,

$$F(r_i) = E.$$

In particular, $\{1, r_i, r_i^2, r_i^3\}$ is an F-basis of E at every round.

Proof. Since $\tau \notin F$, one has $\tau \neq 0$. If $1 - k\tau = 0$ for some admissible integer k, then the image of k in F is nonzero, for otherwise the left-hand side would equal 1. Hence k is invertible in F, so $\tau = k^{-1} \in F$, contradicting $F(\tau) = E$. Therefore the displayed closed forms have no poles.

The formulas themselves follow by induction on k. The base cases $r_1 = \tau$ and $r_2 = \tau^{-1}$ are immediate. If $r_{2k+1} = \tau/(1 - k\tau)$, then

$$r_{2k+2} = \frac{1}{r_{2k+1}} = \frac{1 - k\tau}{\tau}.$$

If $r_{2k+2} = (1 - k\tau)/\tau$, then

$$r_{2k+3} = \frac{1}{r_{2k+2} - 1} = \frac{1}{(1 - (k+1)\tau)/\tau} = \frac{\tau}{1 - (k+1)\tau}.$$

Each r_i is obtained from τ by an invertible Möbius transform with coefficients in F, so $F(r_i) = F(\tau) = E$. Since $[E : F] = 4$, the displayed set is an F-basis. $\square$

For each round i, let

$$m_i(X) = X^4 - \mu_{i,3}X^3 - \mu_{i,2}X^2 - \mu_{i,1}X - \mu_{i,0}$$

be the monic minimal polynomial of r_i over F, so that

$$r_i^4 = \mu_{i,3}r_i^3 + \mu_{i,2}r_i^2 + \mu_{i,1}r_i + \mu_{i,0}.$$

Write

$$\mu_i := (\mu_{i,0}, \mu_{i,1}, \mu_{i,2}, \mu_{i,3}) \in F^4.$$

A live entry at round i is stored as a four-word state

$$s = (a_0, a_1, a_2, a_3) \in F^4,$$

interpreted through a shared denominator $\Delta_i \in E^\times$ as

$$[\![s]\!]_i = \Delta_i^{-1}(a_0 + a_1r_i + a_2r_i^2 + a_3r_i^3).$$

Equivalently, if

$$P_s(Y) := a_0 + a_1Y + a_2Y^2 + a_3Y^3,$$

then $[\![s]\!]_i = \Delta_i^{-1}P_s(r_i)$.

Initialization is scalar:

$$\Delta_1 = 1, \qquad s(v) = (v, 0, 0, 0) \quad (v \in F).$$

The shared denominator evolves by

$$\Delta_{i+1} = \Delta_i r_{i+1}^3.$$

Lemma 2.2 (Closed forms for the shared denominator)**.** *For every admissible integer k, one has*

$$\Delta_{2k} = \tau^{-3}, \qquad \Delta_{2k+1} = (1 - k\tau)^{-3}.$$

Proof. The base cases are $\Delta_1 = 1 = (1 - 0 \cdot \tau)^{-3}$ and

$$\Delta_2 = \Delta_1 r_2^3 = \tau^{-3}.$$

Assume $\Delta_{2k} = \tau^{-3}$. Using Lemma 2.1,

$$\Delta_{2k+1} = \Delta_{2k}r_{2k+1}^3 = \tau^{-3}\left(\frac{\tau}{1 - k\tau}\right)^3 = (1 - k\tau)^{-3}.$$

Then

$$\Delta_{2k+2} = \Delta_{2k+1}r_{2k+2}^3 = (1 - k\tau)^{-3}\left(\frac{1 - k\tau}{\tau}\right)^3 = \tau^{-3}.$$

This proves both formulas by induction. □

3 Exact Local Reciprocal Fold

Fix an admissible round i, and write $r = r_i$, $c = c_i$, and $\mu = \mu_i$. For adjacent states

$$s_0 = (a_0, a_1, a_2, a_3), \qquad s_1 = (b_0, b_1, b_2, b_3),$$

set $\delta_j = b_j - a_j$. Define the intermediate state $u = (u_0, u_1, u_2, u_3)$ by

$$u_0 = a_0 + \mu_0\delta_3, \qquad u_1 = a_1 + \delta_0 + \mu_1\delta_3,$$
$$u_2 = a_2 + \delta_1 + \mu_2\delta_3, \qquad u_3 = a_3 + \delta_2 + \mu_3\delta_3.$$

Equivalently,

$$u = s_0 + S_\mu(s_1 - s_0), \qquad S_\mu = \begin{pmatrix} 0 & 0 & 0 & \mu_0 \\ 1 & 0 & 0 & \mu_1 \\ 0 & 1 & 0 & \mu_2 \\ 0 & 0 & 1 & \mu_3 \end{pmatrix}.$$

The chart transport is

$$R_0(u_0, u_1, u_2, u_3) = (u_3, u_2, u_1, u_0),$$

for odd rounds, and

$$R_1(u_0, u_1, u_2, u_3) = \big(u_3,\ u_2 + 3u_3,\ u_1 + 2u_2 + 3u_3,\ u_0 + u_1 + u_2 + u_3\big),$$

for even rounds.

The Step-A update satisfies the exact identity

$$P_u(r) = P_{s_0}(r) + r\big(P_{s_1}(r) - P_{s_0}(r)\big).$$

In particular, all seed dependence in the local fold is compressed into the coefficient tuple μ_i, and within Step A it enters only through the single scalar δ_3.

Lemma 3.1 (Transport identities)**.** *For every $u \in F^4$,*

$$P_{R_0 u}(Y) = Y^3 P_u(Y^{-1}), \qquad P_{R_1 u}(Y) = Y^3 P_u(1 + Y^{-1}).$$

Consequently, if $r_{i+1} = 1/(r_i - c_i)$ and $\Delta_{i+1} = \Delta_i r_{i+1}^3$, then for $s' = R_{c_i} u$,

$$\Delta_{i+1}^{-1} P_{s'}(r_{i+1}) = \Delta_i^{-1} P_u(r_i).$$

Proof. The displayed polynomial identities are immediate from the definitions of R_0 and R_1. For $c_i = 0$, one has $r_i = r_{i+1}^{-1}$; for $c_i = 1$, one has $r_i = 1 + r_{i+1}^{-1}$. Substitution together with $\Delta_{i+1} = \Delta_i r_{i+1}^3$ yields the claim. □

Theorem 3.2 (Exact reciprocal-chart fold)**.** *Let $u = s_0 + S_{\mu_i}(s_1 - s_0)$ and $s' = R_{c_i} u$. Then*

$$[\![s']\!]_{i+1} = [\![s_0]\!]_i + r_i([\![s_1]\!]_i - [\![s_0]\!]_i).$$

In particular, each physical fold from length $N/2^{i-1}$ to length $N/2^i$ is exact, the live state width stays four over F, and the denominator update is globally shared.

Proof. By the quartic relation for r_i,

$$P_u(r_i) = P_{s_0}(r_i) + r_i\big(P_{s_1}(r_i) - P_{s_0}(r_i)\big).$$

Applying Lemma 3.1 to $s' = R_{c_i} u$ gives

$$\Delta_{i+1}^{-1} P_{s'}(r_{i+1}) = \Delta_i^{-1} P_u(r_i),$$

which is exactly the displayed identity after substituting the definition of $P_u(r_i)$. □

4 Global Structured Evaluator and Reduced Descriptor

The local rule of Theorem 3.2 is linear over F. At round i, each adjacent pair evolves by

$$(s_{2t}, s_{2t+1}) \longmapsto A_i s_{2t} + B_i s_{2t+1},$$

where

$$A_i = R_{c_i}(I - S_{\mu_i}), \qquad B_i = R_{c_i} S_{\mu_i}.$$

The first round simplifies further because the leaves are scalar states.

Lemma 4.1 (Seed-independence of the first round)**.** *For any $v_0, v_1 \in F$,*

$$R_0\big(s(v_0) + S_{\mu_1}(s(v_1) - s(v_0))\big) = (0, 0, v_1 - v_0, v_0),$$

independently of μ_1.

Proof. Since $s(v_1) - s(v_0) = (v_1 - v_0, 0, 0, 0)$, one has

$$S_{\mu_1}(s(v_1) - s(v_0)) = (0, v_1 - v_0, 0, 0).$$

Thus

$$s(v_0) + S_{\mu_1}(s(v_1) - s(v_0)) = (v_0, v_1 - v_0, 0, 0),$$

and applying R_0 yields the claim. □

From this point onward, whenever complexity is mentioned we work in the finite-field specialization

$$F = \mathbb{F}_q, \qquad E = \mathbb{F}_{q^4}, \qquad \mathrm{char}(F) > 3,$$

and count arithmetic operations over F in the unit-cost model. Public descriptors are encoded relative to a fixed ambient F-basis of E, while live fold states continue to use the moving round basis $\{1, r_i, r_i^2, r_i^3\}$.

Define the public descriptors

$$d^{\mathrm{seed}}_{\tau,n} := (n, \tau), \qquad d^{\mathrm{fold}}_{\tau,n} := (\mu_2, \ldots, \mu_n) \in F^{4(n-1)},$$

where for $n = 1$ the fold descriptor is the empty tuple, and

$$d^{\mathrm{rec}}_{\tau,n} := (r_{n+1}, \Delta_{n+1}) \in E \times E^{\times}.$$

The fold descriptor describes the evaluator on F-valued leaves; the reconstruction descriptor is only needed to recover the terminal E-valued quantity represented by the final state.

For the seed-independent first round, define the fixed branch vectors

$$\nu^{(0)} := (0, 0, -1, 1)^{\top}, \qquad \nu^{(1)} := (0, 0, 1, 0)^{\top}.$$

For rounds $i \geq 2$, keep the branch matrices

$$M_i^{(0)} := A_i, \qquad M_i^{(1)} := B_i.$$

Theorem 4.2 (Structured evaluator and reduced descriptor)**.** *For every $\tau \in \Omega^{\mathrm{alg}}$, the reciprocal fold engine defines a public F-linear map*

$$T_{\tau,n} : F^N \to F^4.$$

Moreover:

1. *There is a deterministic expansion algorithm*

$$\mathsf{Expand} : (n, \tau) \longmapsto (d^{\mathrm{fold}}_{\tau,n}, d^{\mathrm{rec}}_{\tau,n})$$

running in $O(n)$ base-field operations.

2. *The reduced fold descriptor has size exactly $4(n-1)$ base-field words. Relative to a fixed ambient F-basis of E, the reconstruction descriptor has size exactly 8 base-field words.*

3. *For each $j = \sum_{i=1}^{n} b_i 2^{i-1}$, the j-th column of $T_{\tau,n}$ is*

$$T_{\tau,n}(e_j^{(N)}) = \begin{cases} \nu^{(b_1)}, & n = 1, \\ M_n^{(b_n)} \cdots M_2^{(b_2)} \nu^{(b_1)}, & n \geq 2, \end{cases}$$

where $e_j^{(N)}$ is the j-th standard basis vector of F^N. Hence any coefficient of the matrix of $T_{\tau,n}$ is computable in $O(n)$ time from $d^{\mathrm{fold}}_{\tau,n}$.

4. *Given $d^{\mathrm{fold}}_{\tau,n}$ and $x \in F^N$, the physical fold engine evaluates $T_{\tau,n}(x)$ in $O(N)$ base-field operations using $O(1)$ field words per live entry.*

Proof. Linearity follows because each round is an explicit F-linear block map and the initialization is linear. For expansion, represent E relative to a fixed ambient F-basis and compute the minimal polynomial of τ by constant-size linear algebra; this costs $O(1)$ because the extension degree is fixed at 4. Proposition A.1 then updates the coefficient tuple round by round in $O(1)$ time per round, and the orbit point r_{i+1} together with the shared denominator Δ_{i+1} also update in $O(1)$ time per round. Thus Expand runs in $O(n)$ time. The output keeps only $\mu_2, \ldots, \mu_n$ because Lemma 4.1 shows that the first round on scalar leaves is seed-independent. Item (2) is immediate from the descriptor definitions.

For Item (3), the first path bit b_1 selects one of the two fixed round-1 branch vectors $\nu^{(0)}$ or $\nu^{(1)}$. Each later bit b_i selects the left or right branch matrix $M_i^{(0)}$ or $M_i^{(1)}$. Composing these choices gives the displayed column formula. Each step is a constant-size matrix-vector multiplication, so the cost is $O(n)$. Item (4) is exactly the physical fold engine of Theorem 3.2 applied layer by layer to the live array. □

Remark 4.3 (Reduced versus uniform per-round descriptors)**.** *The theorem above states the reduced descriptor size for the evaluator $T_{\tau,n} : F^N \to F^4$ on F-valued leaves. If one instead wants a uniform round-by-round transcript, one may also store μ_1, obtaining a descriptor of size exactly $4n$. This extra first-round tuple is algebraically harmless but not needed for the evaluator itself.*

In particular, there exist unique row vectors

$$\lambda^{(0)}_{\tau,n}, \lambda^{(1)}_{\tau,n}, \lambda^{(2)}_{\tau,n}, \lambda^{(3)}_{\tau,n} \in F^N$$

such that

$$T_{\tau,n}(x) = (\langle \lambda^{(0)}_{\tau,n}, x\rangle, \langle \lambda^{(1)}_{\tau,n}, x\rangle, \langle \lambda^{(2)}_{\tau,n}, x\rangle, \langle \lambda^{(3)}_{\tau,n}, x\rangle).$$

The terminal extension-field value represented by the final state is recovered by the public reconstruction map

$$\mathsf{Rec}_{\tau,n}(z_0, z_1, z_2, z_3) := \Delta_{n+1}^{-1}(z_0 + z_1 r_{n+1} + z_2 r_{n+1}^2 + z_3 r_{n+1}^3) \in E.$$

Remark 4.4 (Why quartic is the whole paper)**.** *The quartic case is singled out because it already yields a nontrivial constant-width evaluator with fully explicit local transport, a closed-form coefficient update, and exact arithmetic constants. Higher-degree analogues are a natural next direction.*

Remark 4.5 (Materializing explicit rows)**.** *The full $4 \times N$ matrix of $T_{\tau,n}$ can be materialized in $O(N)$ time by reverse propagation through the depth-n binary circuit. This is routine once the branch vectors and branch matrices are known, but it matters later when one asks whether a downstream layer can consume the structured descriptor directly or only explicit rows.*

5 Worked Example: $N = 4$

The case $N = 4 = 2^2$ is the first nontrivial instance and is worth writing out explicitly. Let

$$x = (x_0, x_1, x_2, x_3) \in F^4.$$

At round 1, the two leaf pairs are (x_0, x_1) and (x_2, x_3). By Lemma 4.1, the first round is seed-independent:

$$s_{01}^{(1)} = (0, 0, x_1 - x_0, x_0), \qquad s_{23}^{(1)} = (0, 0, x_3 - x_2, x_2).$$

In particular, μ_1 does not appear anywhere in the evaluator for $N = 4$.

Now let $\mu_2 = (\mu_{2,0}, \mu_{2,1}, \mu_{2,2}, \mu_{2,3})$. The single surviving pair at round 2 is

$$s_0 = (0, 0, x_1 - x_0, x_0), \qquad s_1 = (0, 0, x_3 - x_2, x_2).$$

Hence

$$\delta_0 = \delta_1 = 0, \qquad \delta_2 = (x_3 - x_2) - (x_1 - x_0), \qquad \delta_3 = x_2 - x_0,$$

and Step A yields

$$u_0 = \mu_{2,0}(x_2 - x_0), \qquad u_1 = \mu_{2,1}(x_2 - x_0),$$
$$u_2 = (x_1 - x_0) + \mu_{2,2}(x_2 - x_0), \qquad u_3 = x_0 + \delta_2 + \mu_{2,3}(x_2 - x_0).$$

Applying the even-round transport gives the final state

$$T_{\tau,2}(x) = R_1 u = (z_0, z_1, z_2, z_3),$$

with explicit matrix form

$$T_{\tau,2}(x) = \begin{pmatrix} 2 - \mu_{2,3} & -1 & \mu_{2,3} - 1 & 1 \\ 5 - \mu_{2,2} - 3\mu_{2,3} & -2 & \mu_{2,2} + 3\mu_{2,3} - 3 & 3 \\ 4 - \mu_{2,1} - 2\mu_{2,2} - 3\mu_{2,3} & -1 & \mu_{2,1} + 2\mu_{2,2} + 3\mu_{2,3} - 3 & 3 \\ 1 - \sum_{j=0}^{3} \mu_{2,j} & 0 & \sum_{j=0}^{3} \mu_{2,j} - 1 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

This makes the global linearity completely concrete: the four displayed rows are precisely

$$\lambda_{\tau,2}^{(0)}, \lambda_{\tau,2}^{(1)}, \lambda_{\tau,2}^{(2)}, \lambda_{\tau,2}^{(3)}.$$

The multiplication count also matches the general theorem of the next section: round 1 costs zero multiplications, round 2 costs exactly four, hence

$$M_{\text{canon}}(4) = 4 = 2 \cdot 4 - 4.$$

6 Exact Canonical Multiplication Count

The fold engine admits an exact multiplication count under a simple straight-line implementation convention.

- Leaves are initialized as scalars $(v, 0, 0, 0)$.
- In round 1, scalar inputs are processed by the specialized routine

$$\mathsf{FoldPairScalar}(v_0, v_1) := (0, 0, v_1 - v_0, v_0),$$

  which is exactly the odd-round transport of $(v_0, v_1 - v_0, 0, 0)$ and uses no field multiplication.
- From round 2 onward, the engine uses the generic Step-A update with the four products

$$\mu_0\delta_3, \quad \mu_1\delta_3, \quad \mu_2\delta_3, \quad \mu_3\delta_3,$$

  followed by the public transport R_0 or R_1.
- Additions, subtractions, rewiring, and add-chain realizations of multiplication by the public constants 2 and 3 are free.
- No further runtime sparsity specialization is performed after round 1.

Proposition 6.1 (Exact canonical multiplication count)**.** *Assume $N = 2^n$ with $n \geq 1$. Under the canonical implementation above, the reciprocal fold engine performs exactly*

$$M_{\text{canon}}(N) = 2N - 4$$

base-field multiplications on F-valued leaves.

Here the count refers only to the local fold arithmetic once the public tuples $\mu_2, \ldots, \mu_n$ are already known. It excludes descriptor expansion, terminal reconstruction in E, and any commitment/opening overhead.

Proof. Round 1 contains $N/2$ pairs, and each is processed by $\mathsf{FoldPairScalar}$, so round 1 contributes zero multiplications. Every later pair contributes exactly four multiplications from Step A and none from transport. The number of pairs after round 1 is

$$\sum_{i=2}^{n} \frac{N}{2^i} = \frac{N}{2} - 1.$$

Therefore

$$M_{\text{canon}}(N) = 4\left(\frac{N}{2} - 1\right) = 2N - 4.$$

□

If multiplication by the public constants 2 and 3 is charged as genuine field multiplication, then every even-round transport R_1 can be realized with at most two additional multiplications per surviving pair: one for $3u_3$, and one for $2u_2$, while $2u_3 = (3u_3) - u_3$ is then obtained for free. Summing those additional costs over even rounds gives

$$M_{\text{cons}}(N) \leq M_{\text{canon}}(N) + 2 \sum_{\substack{2 \leq i \leq n \\ i \text{ even}}} \frac{N}{2^i} < 2N - 4 + \frac{2N}{3} = \frac{8}{3}N - 4.$$

Proposition 6.1 captures a structural property of the reciprocal chart: all seed dependence is concentrated into the single scalar δ_3 per surviving pair, while the chart transport itself is multiplication-free under the canonical model.

Direct Fixed-Basis Baseline

To calibrate Proposition 6.1, consider the most direct ambient-basis implementation of the same logical fold: each live value is stored in a fixed F-basis of E, and each surviving pair is updated by

$$w' = w_0 + r_i(w_1 - w_0),$$

where $r_i \in E$ is the public round challenge. Write $m_{F,\mathrm{pub}}$ for the number of base-field multiplications needed to compute $r\delta$ when $r \in E$ is a generic public constant and $\delta \in F$, and write $m_{E,\mathrm{pub}}$ for the corresponding count when $\delta \in E$, under the chosen quartic multiplication routine.

Proposition 6.2 (Direct fixed-basis baseline)**.** *Under the same multiplication-dominant model as Proposition 6.1, the direct fixed-basis update performs*

$$M_{\mathrm{dir}}(N) = \frac{N}{2}\, m_{F,\mathrm{pub}} + \left(\frac{N}{2} - 1\right) m_{E,\mathrm{pub}}$$

base-field multiplications on F-valued leaves. Consequently,

$$\frac{M_{\mathrm{dir}}(N)}{M_{\mathrm{canon}}(N)} \longrightarrow \frac{m_{F,\mathrm{pub}} + m_{E,\mathrm{pub}}}{4} \qquad \text{as } N \to \infty.$$

Proof. In round 1, each pair begins from F-valued leaves, so $w_1 - w_0 \in F$ and the direct update costs $m_{F,\mathrm{pub}}$ per pair. There are $N/2$ such pairs. From round 2 onward, each live state is a general element of E, so each surviving pair costs $m_{E,\mathrm{pub}}$, and the number of such pairs is

$$\sum_{i=2}^{n} \frac{N}{2^i} = \frac{N}{2} - 1.$$

Summing the two contributions gives the formula, and the asymptotic ratio follows from Proposition 6.1. □

Remark 6.3 (About concrete baseline constants)**.** *The symbolic parameters $m_{F,\mathrm{pub}}$ and $m_{E,\mathrm{pub}}$ absorb the choice of ambient basis and quartic multiplication routine. Proposition 6.2 is the stable statement; any concrete numerical instantiation should be stated relative to an explicitly fixed ambient representation.*

7 Explicit-Row Interface Consequence

Theorems 3.2 and 4.2 identify a concrete backend object for later cryptographic layers: the seed-specialized evaluator $T_{\tau,n}$ together with its reduced descriptor $d^{\mathrm{fold}}_{\tau,n}$. A downstream commitment/opening layer can interact with this backend in two natural ways. It can consume the structured descriptor directly, as in systems designed for structured linear families [5, 8, 1, 9], or it can first materialize the four explicit rows of $T_{\tau,n}$.

The next proposition formalizes the cost of the second choice.

Proposition 7.1 (Explicit-row interfaces destroy descriptor succinctness)**.** *Fix $\tau \in \Omega^{\mathrm{alg}}$. Let a downstream opening layer represent the k-th output coordinate of $T_{\tau,n}$ by an explicit row vector*

$$w^{(k)}_{\tau,n} \in F^N$$

with the correctness property

$$\langle w_{\tau,n}^{(k)}, x\rangle = \big(T_{\tau,n}(x)\big)_k \qquad \textit{for all } x \in F^N.$$

Then necessarily

$$w_{\tau,n}^{(k)} = \lambda_{\tau,n}^{(k)}.$$

Hence any plain explicit-row specialization to τ exposes exactly $4N$ base-field coefficients, whereas the reduced fold descriptor has size $4(n-1)$ (or $4n$ in the uniform per-round encoding of Remark 4.3).

Proof. By definition of the row vectors $\lambda_{\tau,n}^{(k)}$, one has

$$\langle \lambda_{\tau,n}^{(k)}, x\rangle = \big(T_{\tau,n}(x)\big)_k \qquad \forall x \in F^N.$$

If $w_{\tau,n}^{(k)}$ is another correct explicit row, then

$$\langle w_{\tau,n}^{(k)} - \lambda_{\tau,n}^{(k)}, x\rangle = 0 \qquad \forall x \in F^N.$$

Taking $x = e_j^{(N)}$ for each standard basis vector shows

$$w_{\tau,n}^{(k)}[j] = \lambda_{\tau,n}^{(k)}[j] \qquad \forall j,$$

so the two rows are equal. □

Proposition 7.1 is an interface statement, not a hardness statement: by Remark 4.5, the rows can be materialized in linear time from the structured circuit of Theorem 4.2. A row-only interface inevitably replaces a logarithmic-size seed specialization by $4N$ explicit coefficients. Any downstream layer that wants to preserve the $O(n)$-word specialization must therefore consume the structured reciprocal descriptor itself, or something equivalent to it, rather than only explicit row vectors.

8 Concluding Remarks

From the perspective of protocol design, the separation established in Proposition 7.1 provides a clear interface contract. A commitment/opening mechanism can target the evaluator family with exact width, exact descriptor size, and exact fold arithmetic [5, 8, 1, 9]. Likewise, a Fiat–Shamir or correlated-challenge analysis can treat the reciprocal chart as a fixed algebraic primitive rather than rebuilding its local identities inside the security proof [7, 3]. The appendix adds a minimal information-theoretic oracle-model soundness statement for the evaluator relation.

A Reciprocal-Shift Coefficient Update

Proposition A.1 (Reciprocal-shift minimal-polynomial update)**.** *Let*

$$m_i(X) = X^4 - \mu_{i,3}X^3 - \mu_{i,2}X^2 - \mu_{i,1}X - \mu_{i,0}$$

be the monic minimal polynomial of r_i, fix $c \in \{0,1\}$, and define

$$D_c(\mu_i) := m_i(c) = c^4 - \mu_{i,3}c^3 - \mu_{i,2}c^2 - \mu_{i,1}c - \mu_{i,0}.$$

Then $D_c(\mu_i) \neq 0$, and the next orbit point

$$r_{i+1} = \frac{1}{r_i - c}$$

has monic minimal polynomial

$$m_{i+1}(Y) = \frac{Y^4 m_i(c + Y^{-1})}{D_c(\mu_i)}.$$

Equivalently,

$$m_{i+1}(Y) = Y^4 - \mu_{i+1,3}Y^3 - \mu_{i+1,2}Y^2 - \mu_{i+1,1}Y - \mu_{i+1,0},$$

where

$$\mu_{i+1,0} = -D_c(\mu_i)^{-1}, \qquad \mu_{i+1,1} = (\mu_{i,3} - 4c)D_c(\mu_i)^{-1},$$
$$\mu_{i+1,2} = (\mu_{i,2} + 3c\mu_{i,3} - 6c^2)D_c(\mu_i)^{-1},$$
$$\mu_{i+1,3} = (\mu_{i,1} + 2c\mu_{i,2} + 3c^2\mu_{i,3} - 4c^3)D_c(\mu_i)^{-1}.$$

Proof. Because m_i is irreducible of degree 4 over F, it has no root in F. Hence $D_c(\mu_i) = m_i(c) \neq 0$ for every $c \in F$, in particular for $c \in \{0, 1\}$.

Define

$$\widetilde{m}(Y) := \frac{Y^4 m_i(c + Y^{-1})}{D_c(\mu_i)} \in F[Y].$$

Substituting $Y = r_{i+1}$ gives

$$\widetilde{m}(r_{i+1}) = \frac{r_{i+1}^4 m_i(c + r_{i+1}^{-1})}{D_c(\mu_i)} = \frac{r_{i+1}^4 m_i(r_i)}{D_c(\mu_i)} = 0,$$

so $\widetilde{m}$ annihilates r_{i+1}. Moreover, the coefficient of Y^4 in $Y^4 m_i(c + Y^{-1})$ is precisely $m_i(c) = D_c(\mu_i)$, so $\widetilde{m}$ is monic of degree 4.

The Möbius relations

$$r_i = c + r_{i+1}^{-1}, \qquad r_{i+1} = (r_i - c)^{-1}$$

show that $r_i \in F(r_{i+1})$ and $r_{i+1} \in F(r_i)$. Therefore

$$F(r_{i+1}) = F(r_i) = E,$$

so the minimal polynomial of r_{i+1} over F has degree 4. Since $\widetilde{m}$ is already a monic degree-4 polynomial annihilating r_{i+1}, it must equal the minimal polynomial m_{i+1}.

It remains to expand. A direct computation gives

$$Y^4 m_i(c + Y^{-1}) = D_c(\mu_i)Y^4 + (4c^3 - 3c^2\mu_{i,3} - 2c\mu_{i,2} - \mu_{i,1})Y^3$$
$$+(6c^2 - 3c\mu_{i,3} - \mu_{i,2})Y^2 + (4c - \mu_{i,3})Y + 1.$$

Dividing by $D_c(\mu_i)$ and comparing with the monic quartic form

$$Y^4 - \mu_{i+1,3}Y^3 - \mu_{i+1,2}Y^2 - \mu_{i+1,1}Y - \mu_{i+1,0}$$

yields exactly the displayed coefficient formulas. □

B Information-Theoretic Oracle-Model Soundness

This section establishes an information-theoretic soundness theorem for the evaluator relation

$$\mathcal{R}_{\tau,n} := \{(x, y) \in F^N \times F^4 : y = T_{\tau,n}(x)\}.$$

The main result is the following:

> if the prover's claimed output $y \in F^4$ is fixed before seeing a random scalarization vector $\rho \in F^4$, if the sumcheck phase uses fresh independent verifier challenges in F, and if the verifier has exact access to $\widetilde{x}(z)$ at the final sampled point z, then a false relation claim $y \neq T_{\tau,n}(x)$ is accepted with probability at most $(2n + 1)/q$.

The same section also explains why the more naive deterministic-orbit transcript is perfectly unsound. The stated error bound is meaningful in the natural regime $q \gg n$; its leading $1/q$ term is the unavoidable collision loss of a single random scalarization.

B.1 Setup

Throughout this section, the protocol-level probability bounds use only the finite-field assumptions

$$F = \mathbb{F}_q, \qquad N = 2^n, \quad n \geq 1.$$

To instantiate the specific reciprocal evaluator from the quartic note, we additionally work with

$$E = \mathbb{F}_{q^4}, \qquad \mathrm{char}(F) > 3,$$

fix a seed $\tau \in E$ with $F(\tau) = E$, and let

$$r_1 = \tau, \qquad r_{i+1} = \frac{1}{r_i - c_i}, \qquad c_i = \begin{cases} 0, & i \text{ odd}, \\ 1, & i \text{ even}. \end{cases}$$

As proved in the algebraic part of the note, every orbit point satisfies $F(r_i) = E$; in particular,

$$r_i \notin F, \qquad \text{hence} \qquad r_i \notin \{0, 1\}$$

for every round i.

We use the reduced-descriptor notation already established for the evaluator

$$T_{\tau,n} : F^N \to F^4.$$

Thus the first round is represented by the fixed branch vectors

$$\nu^{(0)} = (0, 0, -1, 1)^\top, \qquad \nu^{(1)} = (0, 0, 1, 0)^\top,$$

and every later round $i \geq 2$ is represented by the branch matrices

$$M_i^{(0)}, \qquad M_i^{(1)}.$$

For a bit string $b = (b_1, \ldots, b_n) \in \{0, 1\}^n$, define

$$w_{\tau,\rho}(b) := \begin{cases} \rho^\top \nu^{(b_1)}, & n = 1, \\ \rho^\top M_n^{(b_n)} \cdots M_2^{(b_2)} \nu^{(b_1)}, & n \geq 2, \end{cases}$$

for every scalarization vector $\rho \in F^4$.

For $u = (u_1, \ldots, u_n) \in F^n$, define the affine interpolants

$$\overline{\nu}(u_1) := (1 - u_1)\nu^{(0)} + u_1\nu^{(1)},$$

and, for $i \geq 2$,

$$\overline{M}_i(u_i) := (1 - u_i)M_i^{(0)} + u_i M_i^{(1)}.$$

Then define the associated multilinear weight polynomial by

$$\widetilde{w}_{\tau,\rho}(u) := \begin{cases} \rho^\top \overline{\nu}(u_1), & n = 1, \\ \rho^\top \overline{M}_n(u_n) \cdots \overline{M}_2(u_2)\overline{\nu}(u_1), & n \geq 2. \end{cases}$$

Finally, identify every vector $x \in F^N$ with a function

$$x : \{0,1\}^n \to F, \qquad b \longmapsto x_b,$$

and let

$$\widetilde{x} \in F[U_1, \ldots, U_n]$$

denote its multilinear extension.

Lemma B.1 (Descriptor interpolation)**.** *For every $b \in \{0,1\}^n$,*

$$\widetilde{w}_{\tau,\rho}(b) = w_{\tau,\rho}(b).$$

Moreover, $\widetilde{w}_{\tau,\rho}$ is multilinear in each variable, and for every point $u \in F^n$ its value $\widetilde{w}_{\tau,\rho}(u)$ is computable from $(d^{\mathrm{fold}}_{\tau,n}, \rho)$ in $O(n)$ base-field operations.

Proof. At a Boolean point $b \in \{0,1\}^n$, every affine factor specializes to the branch chosen by the corresponding bit:

$$\overline{\nu}(b_1) = \nu^{(b_1)}, \qquad \overline{M}_i(b_i) = M_i^{(b_i)} \quad (i \geq 2).$$

Substituting these identities into the definition of $\widetilde{w}_{\tau,\rho}$ gives exactly the branch-product formula for $w_{\tau,\rho}(b)$.

Multilinearity is immediate because each factor depends affinely on its own variable and is independent of the others. For evaluation cost, once $u \in F^n$ and $\rho \in F^4$ are fixed, computing $\widetilde{w}_{\tau,\rho}(u)$ amounts to forming a product of $n-1$ constant-size matrices with a constant-size vector, followed by a final dot product with ρ. Every step costs $O(1)$ field operations, so the total cost is $O(n)$. □

Lemma B.2 (Scalarization identity)**.** *For every $x \in F^N$ and every $\rho \in F^4$,*

$$\langle \rho, T_{\tau,n}(x) \rangle = \sum_{b \in \{0,1\}^n} x_b \, w_{\tau,\rho}(b) = \sum_{b \in \{0,1\}^n} \widetilde{x}(b) \, \widetilde{w}_{\tau,\rho}(b).$$

Proof. Write

$$x = \sum_{b \in \{0,1\}^n} x_b e_b,$$

where $e_b \in F^N$ denotes the standard basis vector indexed by b. By linearity of $T_{\tau,n}$,

$$T_{\tau,n}(x) = \sum_{b \in \{0,1\}^n} x_b T_{\tau,n}(e_b).$$

Applying $\rho^\top$ yields

$$\langle \rho, T_{\tau,n}(x) \rangle = \sum_{b \in \{0,1\}^n} x_b \, \rho^\top T_{\tau,n}(e_b) = \sum_{b \in \{0,1\}^n} x_b \, w_{\tau,\rho}(b).$$

The second equality follows because $\widetilde{x}(b) = x_b$ at Boolean points and because Lemma B.1 gives $\widetilde{w}_{\tau,\rho}(b) = w_{\tau,\rho}(b)$. □

B.2 The deterministic-orbit transcript is perfectly unsound

Fix $x \in F^N$ and $\rho \in F^4$, and define

$$H_{\tau,\rho}(u) := \widetilde{x}(u)\, \widetilde{w}_{\tau,\rho}(u) \in F[U_1, \ldots, U_n].$$

Because both factors are multilinear, $H_{\tau,\rho}$ has individual degree at most 2 in every variable. Since its coefficients lie in $F \subseteq E$, it may also be evaluated on points of E^n.

Consider the following deterministic-orbit protocol for the claim

$$\sum_{b \in \{0,1\}^n} H_{\tau,\rho}(b) = \alpha_0.$$

The verifier uses the public orbit points $r_1, \ldots, r_n \in E$ as the round challenges. At round i, the prover sends a polynomial $g_i(T) \in E[T]$ of degree at most 2 satisfying

$$g_i(0) + g_i(1) = \alpha_{i-1},$$

and the verifier sets

$$\alpha_i := g_i(r_i).$$

At the end, the verifier accepts if and only if

$$\alpha_n = H_{\tau,\rho}(r_1, \ldots, r_n).$$

Theorem B.3 (Perfect unsoundness of the deterministic-orbit transcript)**.** *For every false initial claim $\alpha_0 \in E$, there exists a cheating prover that makes the deterministic-orbit protocol above accept with probability* 1*.*

Proof. Fix any sequence of target intermediate claims

$$\alpha_1, \ldots, \alpha_{n-1} \in E, \qquad \alpha_n := H_{\tau,\rho}(r_1, \ldots, r_n).$$

It is enough to show that, for every round i, every prescribed pair $(\alpha_{i-1}, \alpha_i) \in E^2$ can be realized by some degree-≤ 2 polynomial $g_i(T)$ satisfying

$$g_i(0) + g_i(1) = \alpha_{i-1}, \qquad g_i(r_i) = \alpha_i.$$

Consider a polynomial of the form

$$g_i(T) = a_i T^2 + b_i T.$$

The two required conditions become the linear system

$$a_i + b_i = \alpha_{i-1}, \qquad a_i r_i^2 + b_i r_i = \alpha_i.$$

Its determinant is

$$\det \begin{pmatrix} 1 & 1 \\ r_i^2 & r_i \end{pmatrix} = r_i - r_i^2 = r_i(1 - r_i),$$

which is nonzero because $r_i \notin \{0, 1\}$. Therefore the system has a unique solution $(a_i, b_i) \in E^2$. Hence every round constraint can be met exactly.

By construction, the final claim is chosen to equal the verifier's terminal check value. Therefore every round check and the final check pass, even though the initial claim α_0 was false. The verifier accepts unconditionally. □

Remark B.4. *The pathology above does not depend on any special quartic identity beyond the fact that the reused challenges satisfy $r_i \notin \{0, 1\}$. The quartic reciprocal orbit simply provides a concrete deterministic family of such non-Boolean challenges.*

B.3 The idealized oracle-model protocol

The positive result concerns a *different* protocol. Here the prover's claimed output is fixed before scalarization, the sumcheck phase uses fresh random verifier challenges in F, and the verifier only needs the exact value $\widetilde{x}(z)$ at the final sampled point $z \in F^n$.

For a fixed input $x \in F^N$, the protocol is:

1. The prover outputs a claimed evaluator output
$$y \in F^4.$$
This first message is sent before the verifier reveals ρ.

2. The verifier samples
$$\rho \leftarrow F^4$$
uniformly at random and sets
$$\beta_0 := \langle \rho, y \rangle.$$

3. The verifier defines
$$H_{\tau,\rho}(u) := \widetilde{x}(u)\, \widetilde{w}_{\tau,\rho}(u).$$

4. The prover and verifier run the standard n-round sumcheck protocol for the claim
$$\sum_{b \in \{0,1\}^n} H_{\tau,\rho}(b) = \beta_0.$$
At round i, after seeing ρ and the previous verifier challenges $z_1, \ldots, z_{i-1}$, the prover sends a polynomial $h_i(T) \in F[T]$. The verifier checks that $\deg h_i \leq 2$ and that
$$h_1(0) + h_1(1) = \beta_0$$
at the first round, and
$$h_i(0) + h_i(1) = h_{i-1}(z_{i-1})$$
for each later round. The verifier then samples
$$z_i \leftarrow F$$
uniformly and independently.

5. At the final point $z = (z_1, \ldots, z_n) \in F^n$, the verifier queries the exact value $\widetilde{x}(z)$ and checks
$$h_n(z_n) = H_{\tau,\rho}(z) = \widetilde{x}(z)\, \widetilde{w}_{\tau,\rho}(z).$$

The degree check $\deg h_i \leq 2$ is to be understood in the usual coefficient-representation sense: the prover sends h_i by its coefficients in F.

Lemma B.5 (Standard degree-2 sumcheck soundness)**.** *Let*
$$H \in F[U_1, \ldots, U_n]$$
be any polynomial of individual degree at most 2 *in every variable. Consider the standard* n*-round sumcheck protocol for the claim*
$$\sum_{b \in \{0,1\}^n} H(b) = \beta_0,$$

with verifier checks $\deg h_i \le 2$ *at every round and with exact final-point access to* $H(z_1, \ldots, z_n)$*. Then every adaptive cheating prover starting from a false initial claim*

$$\beta_0 \neq \sum_{b \in \{0,1\}^n} H(b)$$

is accepted with probability at most $2n/q$*.*

Proof. We argue by induction on n.

For $n = 1$, the verifier checks

$$h_1(0) + h_1(1) = \beta_0$$

and finally checks $h_1(z_1) = H(z_1)$. Since the initial claim is false,

$$\beta_0 \neq H(0) + H(1).$$

Therefore any first-round polynomial h_1 satisfying the verifier's consistency equation must differ from the true polynomial $H(T)$. Hence

$$q_1(T) := h_1(T) - H(T)$$

is a nonzero polynomial of degree at most 2. Acceptance implies $q_1(z_1) = 0$, which occurs with probability at most $2/q$.

Assume now that the statement holds for $(n-1)$ variables, and consider an n-variable instance. Define the true first-round partial-sum polynomial

$$p_1(T) := \sum_{b_2, \ldots, b_n \in \{0,1\}^{n-1}} H(T, b_2, \ldots, b_n).$$

Because H has individual degree at most 2, the univariate polynomial p_1 has degree at most 2. Since the initial claim is false,

$$p_1(0) + p_1(1) = \sum_{b \in \{0,1\}^n} H(b) \neq \beta_0.$$

Thus any first-round prover message h_1 satisfying

$$h_1(0) + h_1(1) = \beta_0$$

must differ from p_1. Let

$$q_1(T) := h_1(T) - p_1(T),$$

so q_1 is a nonzero polynomial of degree at most 2.

The verifier next samples $z_1 \leftarrow F$. If $q_1(z_1) = 0$, then the residual claim may become true, and we upper-bound the remaining acceptance probability by 1. Since q_1 has at most 2 roots,

$$\Pr[q_1(z_1) = 0] \le \frac{2}{q}.$$

On the complementary event $q_1(z_1) \neq 0$, the residual claim for the specialized polynomial

$$H(z_1, U_2, \ldots, U_n)$$

is still false, because the prover's carried-forward claim is $h_1(z_1)$ while the true carried-forward claim is $p_1(z_1)$. The continuation of the prover is an adaptive adversary for a standard $(n-1)$-round

sumcheck instance whose target polynomial still has individual degree at most 2. By the induction hypothesis, conditioned on $q_1(z_1) \neq 0$, the remaining acceptance probability is at most $2(n-1)/q$.

Combining the two cases gives

$$\Pr[\text{accept}] \leq \Pr[q_1(z_1) = 0] \cdot 1 + \Pr[q_1(z_1) \neq 0] \cdot \frac{2(n-1)}{q} \leq \frac{2}{q} + \frac{2(n-1)}{q} = \frac{2n}{q}.$$

This completes the induction. □

Proposition B.6 (Why y must be fixed before scalarization)**.** *If the prover is allowed to choose y after seeing $\rho \in F^4$, then the scalarization check by itself becomes vacuous: for every $x \in F^N$ and every $\rho \in F^4$, there exists a vector*

$$y \neq T_{\tau,n}(x)$$

such that

$$\langle \rho, y \rangle = \langle \rho, T_{\tau,n}(x) \rangle.$$

Proof. Choose any nonzero vector

$$\delta \in \{v \in F^4 : \langle \rho, v \rangle = 0\}.$$

Such a nonzero vector always exists: if $\rho = 0$, the orthogonal subspace is all of F^4, and if $\rho \neq 0$, it is a 3-dimensional hyperplane. Set

$$y := T_{\tau,n}(x) + \delta.$$

Then $y \neq T_{\tau,n}(x)$ but

$$\langle \rho, y \rangle = \langle \rho, T_{\tau,n}(x) \rangle + \langle \rho, \delta \rangle = \langle \rho, T_{\tau,n}(x) \rangle.$$

Therefore no one-shot scalarization test can certify the full F^4-valued relation unless y is fixed before ρ is sampled. □

Proposition B.7 (Fresh scalarization alone does not repair the deterministic-orbit protocol)**.** *Suppose the verifier samples $\rho \leftarrow F^4$ uniformly at random but still reuses the public orbit points $r_1, \ldots, r_n$ as the round challenges. Then the resulting protocol remains perfectly unsound.*

Proof. Fix any false output claim $y \neq T_{\tau,n}(x)$, and let

$$\Delta := y - T_{\tau,n}(x) \neq 0.$$

For each realized value of ρ, there are two cases.

If $\langle \rho, \Delta \rangle = 0$, then the scalarized initial claim

$$\langle \rho, y \rangle = \langle \rho, T_{\tau,n}(x) \rangle$$

is actually true. In this case an honest deterministic-orbit transcript is accepted.

If instead $\langle \rho, \Delta \rangle \neq 0$, then the scalarized initial claim is false. But Theorem B.3 shows that, for the deterministic-orbit protocol, the prover can still force acceptance with probability 1.

Thus for every realization of ρ there exists an accepting prover strategy, so fresh scalarization without fresh verifier challenges does not restore soundness. □

Theorem B.8 (Information-theoretic oracle-model one-shot soundness)**.** *Consider the idealized protocol above. Let $P^\star$ be any possibly randomized prover that, on input $x \in F^N$, first outputs a vector $y \in F^4$ before seeing ρ, and then adaptively participates in the sumcheck phase after seeing ρ and the verifier challenges $z_1, \ldots, z_{i-1}$. Then*

$$\Pr\left[\textit{verifier accepts and } y \neq T_{\tau,n}(x)\right] \leq \min\left\{1, \frac{2n+1}{q}\right\}.$$

The probability is taken over the internal randomness of $P^\star$ together with the verifier randomness $\rho, z_1, \ldots, z_n$.

Proof. Let ω denote the internal random coins of $P^\star$, and condition on an arbitrary fixed value of ω. Under this conditioning the prover becomes deterministic. Let $y = y(\omega)$ be its first message, and define

$$\Delta := y - T_{\tau,n}(x) \in F^4.$$

If $\Delta = 0$, then the bad event

$$\{\text{verifier accepts and } y \neq T_{\tau,n}(x)\}$$

is impossible, so there is nothing to prove. Assume from now on that $\Delta \neq 0$.

Since y is fixed before ρ is sampled, the vector Δ is independent of ρ. For nonzero Δ, the map

$$\rho \longmapsto \langle \rho, \Delta \rangle$$

is a nonzero linear functional on the 4-dimensional F-vector space F^4, and therefore its kernel has size exactly q^3. Consequently,

$$\Pr_\rho\left[\langle \rho, \Delta \rangle = 0\right] = \frac{1}{q}.$$

Let

$$\mathsf{Coll} := \{\langle \rho, \Delta \rangle = 0\}.$$

On the complementary event $\neg\mathsf{Coll}$, we have

$$\beta_0 = \langle \rho, y \rangle \neq \langle \rho, T_{\tau,n}(x) \rangle.$$

By Lemma B.2,

$$\langle \rho, T_{\tau,n}(x) \rangle = \sum_{b \in \{0,1\}^n} \widetilde{x}(b)\, \widetilde{w}_{\tau,\rho}(b) = \sum_{b \in \{0,1\}^n} H_{\tau,\rho}(b).$$

Hence, conditioned on any fixed non-colliding value of ρ, the prover is trying to make the verifier accept a false sumcheck claim for the polynomial $H_{\tau,\rho}$.

Now fix such a non-colliding value of ρ. By construction, both $\widetilde{x}$ and $\widetilde{w}_{\tau,\rho}$ are multilinear, so their product $H_{\tau,\rho}$ has individual degree at most 2 in each variable. Therefore Lemma B.5 applies and gives

$$\Pr[\text{accept} \mid \omega, \rho, \neg\mathsf{Coll}] \leq \frac{2n}{q}.$$

Using the trivial upper bound 1 on the acceptance probability under Coll, we obtain

$$\Pr[\text{accept and } \Delta \neq 0 \mid \omega] \leq \Pr[\mathsf{Coll} \mid \omega] + \Pr[\text{accept} \mid \omega, \neg\mathsf{Coll}] \leq \frac{1}{q} + \frac{2n}{q} = \frac{2n+1}{q}.$$

Since the same bound holds for every fixed value of ω, averaging over ω preserves the inequality. Finally, clipping by 1 yields the stated bound. □

Remark B.9 (Interpreting the error term)**.** *The bound in Theorem B.8 is informative in the natural parameter regime $q \gg n$. Its two terms have different origins: the additive $1/q$ term is the unavoidable collision probability of the one-shot random scalarization map*

$$v \longmapsto \langle \rho, v \rangle,$$

while the remaining $2n/q$ term comes from the n rounds of degree-2 sumcheck.

Remark B.10 (Exact scope of the theorem)**.** *Theorem B.8 is an information-theoretic oracle-model one-shot relation soundness theorem for pre-ρ-fixed outputs. It establishes the foundational algebraic soundness of the evaluator relation $y = T_{\tau,n}(x)$ once the incorrect deterministic-orbit transcript is discarded. Concrete commitment/opening layers, adaptive multi-query opening, and Fiat–Shamir or correlated-challenge non-interactive analyses build on this statement as subsequent extensions.*

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